

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (Biotechnology) (Credit System) Degree Examinations

Supplementary - April 2023

20BT305 - BIOPROCESS PRINCIPLES AND CALCULATIONS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.
2) Use of **Humidity** chart is permitted.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Enumerate on the different physical quantities with examples and different units in different systems.	10	L*3	1	1
	b) In a reaction $N_2 + 3H_2 \rightarrow 2NH_3$, how much N_2 and H_2 are required to produce 100 kg of NH_3 .	10	L2	1	1
2.	a) State the equation of state and ideal gas law. Find the value of gas constant R in mmHg m^3 /mol K and cal/mol K and also calculate the molar volume of air at 350 K and 1 bar.	10	L3	1	1
	b) Write SI units and dimensions for density, viscosity, surface tension, specific weight, specific volume, pressure, power, heat capacity, force, energy.	10	L2	1	1
Unit – II					
3.	a) 100 kmol/h of 40 mole% of solution of Ethylene dichloride in Toluene is fed to middle of the distillation column. The distillate contains 95 mole% Ethylene dichloride and the bottoms consists of 90 mole% Toluene. What is the rate of flow of each stream?	10	L3	2	2
	b) Soyabean seeds oil is extracted with hexane in a batch extractors. The flaked seeds contain 18.2% oil, 69.5% solid and 12.3% moisture. At the end of the process, cake is separated from hexane oil mixture. The cake analysis yields 0.8% oil, 88.2% solids and 11.0% moisture. Find the percentage recovery of oil. All percentage is by weight.	10	L4	2	2
4.	a) A solution of $NaNO_3$ in water at 40°C contains 48% $NaNO_3$ by weight. Calculate the percentage saturation of this solution. Calculate the weight of $NaNO_3$ that may be crystallized from 1000 kg of this solution by reducing the temperature to 10°C. Calculate the percentage yield. Solubility of $NaNO_3$ at 40°C is 51.4% by weight Solubility of $NaNO_3$ at 10°C is 44.5% by weight.	10	L2	2	2
	b) Explain in general the mixer with neat diagram and material balance. It is required to make 100 kg of 30% NaOH solution by mixing the following liquids; 20% NaOH solution and 36% NaOH solution. Calculate the quantities of the two solutions to be mixed.	10	L3	2	2
Unit – III					
5.	a) A limestone analysis is shown below. $CaCO_3$ = 92.89% $MgCO_3$ = 5.41% Insoluble = 1.70% (All percentage are by mole) (i) How many kilomoles of CaO can be made from 6 kilomoles of this limestone? (ii) How many kilomoles of CO_2 can be recovered per kilomoles of limestone? (iii) How many kilomoles of limestone are needed to make 2 kilomoles of lime?	10	L3	3	1

P.T.O.

- b) Anaerobic digestion of volatile acids by methane bacteria is represented by the equation



The composition of methane bacteria is approximated by the empirical formula $\text{CH}_{1.4}\text{O}_{0.4}\text{N}_{0.2}$. For each kg acetic acid consumed 0.67 kg CO_2 is evolved. How does the yield of methane under these conditions compare with the maximum possible yield?

10 L3 3 2

6. a) A coke contains 85% carbon and 15% non-combustible material by weight. Calculate:
(i) The amount of oxygen theoretically required to burn 120 kg of coke completely.

(ii) The composition of gases in the product stream if 60% excess air is supplied.

10 L3 3 2

- b) Estimate the theoretical growth and product yield coefficients for ethanol fermentation by *S. cerevisiae* as given by the overall reaction.



Yeast yields 2ATP/mol of glucose and

$\text{Y}_{\text{X/ATP}} = 10.5 \text{ g/mol ATP}$.

10 L3 3 2

Unit – IV

7. a) Explain the energy balance in fermentors with a neat flow diagram.
b) Citric acid is manufactured using submerged culture of *Aspergillus niger* in a batch reactor operated at 30°C. Over a period of two days, 2500 kg glucose and 860 kg oxygen are consumed to produce 1500 kg citric acid, 500 kg biomass and other products. Ammonia is used as nitrogen source. Power input to the system by mechanical agitation of the broth is about 15 kW; approximately 100 kg water is evaporated over the culture period. Estimate the cooling requirements.

10 L2 4 2

8. a) Elucidate in detail the heat of formation, reaction and combustion for a biochemical reaction considering a suitable example.
b) Explain the laws of thermodynamic with Carnot cycle and how Carnot cycle has an impact on heat engines and pumps.

10 L2 4 2

10 L3 4 1

Unit – V

9. a) In a tank are 100 liters of brine containing 50 kg of total dissolved salts. Pure water is allowed to run into the tank at the rate of 3 l/min. Brine runs out of the tank at the rate of 2 l/min. The instantaneous concentration in the tank is kept uniform by stirring. How much salt is in the tank at the end of the one hour?

10 L3 5 3

- b) Write the generalized material and energy balance for batch distillation and CSTR.

10 L4 5 2

10. a) A dryer reduces the moisture content of 100kg of a potato product from 80% to 10% moisture. 250kg of steam at 70kPa gauge is used to heat 49,800m³ of air to 80°C and the air is cooled to 71°C in passing through the dryer. Calculate the efficiency of the dryer. The specific heat of potato is 3.43kJ/kg⁻¹°C⁻¹. Assume potato enters at 24°C, which is also the ambient air temperature, and leaves at the same temperature as the exit air.

10 L4 5 2

- b) Elucidate the dryer operation with simultaneous heat and mass balance with the use of psychrometric chart.

10 L3 5 2